

Project Progress Report

(Group – 04)

Week 6: July 17 – July 23, 2025

- **Codebase Refactoring**

- Reorganized code by modularizing major components into separate files.
- Improved readability and maintainability by keeping each file concise.

- **Enhanced Dataset Loader**

- Extended loader to support multiple file formats: .pt, .pth, .pkl, .csv, .json, .txt, .npy, .npz.
- Verified functionality for .pt, .csv, .json, and .txt formats with successful testing.
- Loader dynamically scans a dataset directory, reviews file contents, and automatically selects valid inputs—no reliance on filename conventions.

- **Implemented More Models for Graph-Level Tasks**

- **Graph Matching:** Added a fully functional Graph Matching Network (GMN).
- **Graph Reconstruction:** Implemented Graph Reconstruction Autoencoder (GRAE).
- **Graph Generation:** Implemented GraphRNN for the graph generation task.

- **Model Persistence Layer**

- **Built model save/load functionality:**
 - ❖ Trained models are saved per dataset/task.
 - ❖ On subsequent runs with the same dataset and task, the system automatically loads the existing model instead of retraining.